

Multiple-Choice Questions (MCQs) for Module 1: Audio Signals Fundamentals

Sampling and Analog-to-Digital Conversion

1. What does sampling rate refer to?

- a) Number of amplitude levels
- b) Number of samples per second (Hz)
- c) Duration of the recorded audio
- d) Loudness of the audio

2. According to the Nyquist theorem, if the highest frequency in a signal is 10 kHz, what is the minimum required sampling rate?

- a) 5 kHz
- b) 10 kHz
- c) 15 kHz
- d) 20 kHz

3. Which artifact occurs if the sampling rate is too low?

- a) Distortion
- b) Aliasing
- c) Clipping
- d) Quantization noise

4. The Nyquist frequency is defined as:

- a) Half the sampling rate
- b) Twice the sampling rate
- c) Equal to sampling rate
- d) The highest frequency captured exactly

5. An anti-aliasing filter:

- a) Removes low frequencies
 - b) Removes frequencies above the Nyquist frequency
 - c) Enhances high-frequency content
 - d) Increases bit depth
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Bit Depth and Quantization

6. Bit depth controls:

- a) Sampling rate
- b) Frequency resolution
- c) Amplitude resolution
- d) Maximum frequency

7. How many discrete amplitude levels can 8-bit audio represent?

- a) 8
- b) 16
- c) 128
- d) 256

8. What happens when audio bit depth is very low (e.g., 8-bit)?

- a) Higher dynamic range
- b) Audible quantization noise
- c) Clearer audio
- d) Higher maximum frequency

9. Each additional bit in bit depth roughly adds how much dynamic range?

- a) 1 dB
- b) 3 dB
- c) 6 dB
- d) 12 dB

10. What does quantization error refer to?

- a) Errors in the timing of samples
- b) Errors due to rounding amplitude levels
- c) Errors in microphone placement
- d) Errors in frequency selection

Amplitude and Decibels (dB)

11. In digital audio, 0 dBFS refers to:

- a) Absolute silence
- b) Maximum possible amplitude
- c) Minimum audible amplitude
- d) Reference amplitude for loudness

12. Doubling amplitude corresponds to approximately what dB increase?

- a) 3 dB
- b) 6 dB**
- c) 10 dB
- d) 12 dB

13. A sound at -12 dBFS compared to one at -6 dBFS is:

- a) Louder
- b) The same loudness
- c) Half the amplitude**
- d) Twice the amplitude

14. Decibels are measured on what type of scale?

- a) Linear
- b) Exponential
- c) Logarithmic**
- d) Quadratic

15. Which unit represents digital audio amplitude?

- a) dB SPL**
- b) dBFS**
- c) dBm
- d) Hz

Frequency and Pitch

16. Frequency is measured in:

- a) Decibels
- b) Seconds
- c) Hertz**
- d) Bits

17. A tone of 500 Hz compared to 250 Hz is perceived as:

- a) Lower pitch
- b) Same pitch
- c) Higher pitch**
- d) Same loudness

18. Humans generally hear frequencies in what range?

- a) 20 Hz to 2 kHz
- b) 100 Hz to 10 kHz
- c) 20 Hz to 20 kHz
- d) 1 kHz to 100 kHz

19. The fundamental frequency of speech primarily determines:

- a) Loudness
- b) Pitch
- c) Formant structure
- d) Amplitude resolution

20. What are harmonics in audio signals?

- a) Frequencies below the fundamental frequency
- b) Integer multiples of fundamental frequency
- c) Unrelated background noises
- d) Frequencies above human hearing

Voice Characteristics

21. An adult male's typical fundamental frequency is around:

- a) 60 Hz
- b) 120 Hz
- c) 500 Hz
- d) 1000 Hz

22. Children's voices differ from adults' mainly because:

- a) Their vocal folds vibrate more slowly
- b) Their vocal tract resonances (formants) are lower
- c) Their fundamental and formant frequencies are higher
- d) They produce fewer harmonics

23. Formants are:

- a) Low-frequency noise
- b) High-frequency harmonics only
- c) Vocal tract resonance frequencies
- d) Related to loudness only

24. What most strongly influences voice timbre?

- a) Sampling rate
- b) Fundamental frequency alone
- c) Vocal tract resonances (formants)
- d) Bit depth

25. Which statement about adult female voices compared to male voices is typically true?

- a) Lower pitch and formants
 - b) Higher pitch and formants
 - c) Same pitch, different formants
 - d) Higher pitch, lower formants
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General Audio Processing Concepts

26. Dynamic range in audio signals describes:

- a) Range between the lowest and the highest frequency
- b) Range between the quietest and the loudest sound
- c) Number of bits per second
- d) Frequency spectrum spread

27. Which factor primarily determines the highest frequency captured in digital audio?

- a) Bit depth
- b) Sampling rate
- c) Dynamic range
- d) Audio length

28. Which type of noise is directly introduced by low bit depth?

- a) Aliasing
- b) Quantization noise
- c) White noise
- d) Hum noise

29. In digital audio, if amplitude exceeds 0 dBFS, what happens?

- a) Louder sound without issues
- b) Clipping distortion occurs
- c) Audio improves clarity
- d) Frequency decreases

30. Which describes a higher sample rate's effect within audible range (20 Hz–20 kHz)?

- a) Improved frequency response for human hearing
- b) Improved amplitude resolution
- c) No audible improvement if above Nyquist requirement
- d) Increased loudness